# Solutions and teaching notes for chapter 10

**Teacher use.** The answers below concern the tasks not worked in the lesson. First have students state their method; give graded help before the solution. The durations are estimates to be calibrated in class.

## Diagnostic and remediation

1. $2\pi$. *Action:* Link radius, diameter and perimeter.
2. $90°$. *Action:* Divide a circle into four sectors.
3. $\dfrac{\sqrt{2}}{2}$. *Action:* Rationalise and check the square.
4. $(a;-b)$. *Action:* Draw the two points.
5. $-\dfrac{\pi}{3}$. *Action:* Put over a common denominator.
6. $\dfrac{3}{5}$. *Action:* Identify the reference angle.

## Investigation activity

Equilateral triangle then isosceles right triangle. Pythagoras gives respectively height $\dfrac{\sqrt{3}}{2}$ and equal coordinates $\dfrac{\sqrt{2}}{2}$. The signs are positive in the first quadrant.

## Immediate applications

1. $\dfrac{\pi}{4}$, $\dfrac{2\pi}{3}$; length $\dfrac{3\pi}{2}$ m.
2. $\dfrac{7\pi}{2}$ has the image of $\dfrac{3\pi}{2}$; $-\dfrac{5\pi}{3}$ that of $\dfrac{\pi}{3}$.
3. $\sin^2 x = \dfrac{16}{25}$; the sine is negative, so $-\dfrac{4}{5}$.
4. $(-\dfrac{1}{2};-\dfrac{\sqrt{3}}{2})$ and $(\dfrac{\sqrt{2}}{2};-\dfrac{\sqrt{2}}{2})$.
5. $h = \dfrac{\sqrt{3}}{2}$; new chord $\dfrac{1}{\sqrt{2+\sqrt{3}}} = \sqrt{2-\sqrt{3}}$.

## Exercises: method and results

**Exercise 10.1 – Converting and measuring.** $\dfrac{5\pi}{6}$ and $-\dfrac{\pi}{2}$; arc $\dfrac{4\pi}{3}$ cm. The negative sign indicates an orientation; a geometric length is positive.

**Exercise 10.2 – Reducing angles.** $\dfrac{5\pi}{6}$, $\dfrac{\pi}{4}$, $\pi$. Remove or add multiples of $2\pi$.

**Exercise 10.3 – Exact values.** $(-\frac{1}{2};\frac{\sqrt{3}}{2})$; $(\frac{\sqrt{3}}{2};-\frac{1}{2})$; $(-\frac{\sqrt{2}}{2};-\frac{\sqrt{2}}{2})$, in the form (cosine; sine).

**Exercise 10.4 – Recovering the sign.** $\cos^2 x = 1-\frac{25}{169}=\frac{144}{169}$; cosine negative, so $-\frac{12}{13}$.

**Exercise 10.5 – A wheel.** Advance $r\theta = 0.3(\frac{5\pi}{3}) = \frac{\pi}{2}$ m. This corresponds to the unrolling of the arc; the chord joins two positions on the circle and does not have this length. A material point on the wheel follows a different path relative to the ground.

**Exercise 10.6 – Chord and scalar product.** $A(1;0)$, $B(-\frac{1}{2};\frac{\sqrt{3}}{2})$; $AB^2 = \frac{9}{4}+\frac{3}{4}=3$. Al-Kashi: $1+1-2\cos(\frac{2\pi}{3})=3$. So $AB=\sqrt{3}$.

**Exercise 10.7 – Reading an equation on the circle.** The vertical line with x-coordinate $\frac{1}{2}$ meets the circle at the angles $\frac{\pi}{3}$ and $\frac{5\pi}{3}$. The two points have opposite sines.

**Exercise 10.8 – Delayed review.** $\frac{25\pi}{4} = 6\pi+\frac{\pi}{4}$. Three complete turns do not change the image, so both values are $\frac{\sqrt{2}}{2}$.

**CQFD, exercise 32 p. 220, questions a and b.** a: $\cos(\frac{6\pi}{5}) = \cos(\pi+\frac{\pi}{5}) = -\frac{1+\sqrt{5}}{4}$. b: $\cos(-\frac{\pi}{5}) = \cos(\frac{\pi}{5}) = \frac{1+\sqrt{5}}{4}$, by evenness of the cosine.

**CQFD, exercise 39 p. 220, questions 1, 2, 3 and 5.** 1: $\frac{\pi}{6}+\frac{\pi}{3}=\frac{\pi}{2}$. 2: yes, $\cos(\frac{\pi}{3})=\frac{1}{2}=0.5$ is a decimal. 3: the principal measure in $]-\pi;\pi]$ is $-\frac{5\pi}{6}$. 5: $\sin(\frac{4\pi}{3})=-\frac{\sqrt{3}}{2}$.

## MCQ: answers and reasons

**MCQ 1.**

1. B. Half turn.
2. A. Equilateral triangle.
3. D. Lowest point of the circle.
4. C. Equation of the unit circle.
5. B. Complete turns.

**MCQ 2.**

1. C. Vertical symmetry.

2. A. Isosceles right triangle.
3. D. Three quarters of a turn.
4. B. $r\theta = 6$.
5. C. Negative x-coordinate inside the quadrant.

### True or false

1. False: at 0, the two sides would be 1 and $-1$. The cosine is even.
2. True: a complete turn keeps the image.
3. False: $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ have the same sine but opposite cosines.

**Differentiated teaching.** For a technical block, offer only the first step (domain, factor or reference sample space), then withdraw the help. For an independent student, ask for a second method and the treatment of a limiting case. Orally, allow an initial reformulation in the language the student has mastered, then require a precise mathematical sentence in French.